



Name: Cohort/Batch: Date: Score:

SQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is SQL and what are its core strengths?

Q6 How do you monitor and tune slow queries in SQL?

Q2 How does SQL guarantee consistency and handle transactions?

Q7 What backup and restore strategies does SQL support?

Q3 What index types does SQL support and when do you use each?

Q8 How do you handle schema migrations in SQL?

Q4 How do you design a schema or data model in SQL?

Q9 What security features does SQL provide?

Q5 How does SQL scale horizontally?

Q10 What are common anti-patterns when using SQL in production?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 SQL is a relational, document, key-value, graph,

Mitigation

SQL is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 SQL provides a slow query log, explain/explain

Observability

SQL provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 SQL provides either full ACID transactions or

Primitives

SQL provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 SQL supports logical backups (export SQL or

Automation

SQL supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 SQL offers B-tree, hash, full-text, spatial, or

Hardening

SQL offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in SQL are managed with

Governance

Schema migrations in SQL are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in SQL involves normalizing relations

Gotchas

Schema design in SQL involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 SQL supports role-based access control, encrypted

Assessment

SQL supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 SQL scales via replication (read replicas) for

Federation

SQL scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some SQL implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all columns instead

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.